Bypass-LoRA: 低维循环注意力与 FFN 条件融合 v2

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Abstract

本文记录 Bypass-LoRA v2 计划 V0 至 V2 的结果。研究对象是冻结的 Qwen3-0.6B-Base 旁的低维侧分支。V0 用预先锁定容差的检查验证了侧分支的实现语义（E-0001 修正后 85/85 项、E-0004 130/130 项通过）。V1 在 PIQA+BoolQ 上用不含记忆的 tokenwise proxy 比较八组接口（每组 2M tokens，seed 0）：最好的 concat 接口 C_dual（78.15%）只比加法 F_out（77.80%）高 0.35 pp，未达 1 pp 门槛；两个参数相近的 LoRA（79.20%、79.05%）更高。H-0002 记为 challenged（screening）。V2 研究只保留低维记忆时的单跳读取。事实在上下文中的 F 线正对照训练后通过（calibration EM 99.95%）；按家族校准学习率后，十组 M 线配置（特征 × 读算子 × 融合的 2×2×2 析因，加两个 C_dual 探针，每组 4M tokens）都没有达到运行前锁定的门槛：正常记忆对清零与打乱记忆的 EM 优势最大 +0.6 pp，门槛为 10 pp。H-0003 记为 challenged（screening）。失败定位测量显示，每组训练后在答案位置权重最高的槽位都是第一个写入的槽位，读出向量在独立探针下的答案准确率接近先验；冻结特征在写入与问题位置仍能线性解码答案，并支持有监督寻址。记忆路线按预期价值停止，计划允许的一个探针没有使用，依赖可用读取的 V3/V4 暂停。主要限制是单 seed、筛选级证据，以及失败原因的解释没有被隔离。

1 Introduction

Bypass-LoRA v2 在冻结主干旁加一条低维循环侧分支：从注意力的 Q/K/V 取特征，压缩成低维状态，经记忆读写与循环更新，再通过 FFN 融合写回主干。计划把研究拆成几个阶段：V0 验证实现语义（H-0001）；V1 在没有记忆与循环的 tokenwise proxy 下比较融合接口，检验把信号注入 FFN 的 gate/up 是否优于加法写回（H-0002）；从 V2 起研究丢弃 support 主干 KV、只保留低维记忆时的单跳读取（H-0003），后续还有读算子选择性（H-0004）、多次读取（H-0005）与实用适配（H-0006）。本文报告 V0、V1 与 V2 的结果，以及因 V2 没有产生可用读取而对后续阶段（V3–V5）所作的调整。各假设的陈述、范围与当前状态如下。

- H-0001@1 (supported, active) V0 实现语义与恒等式正确: 新侧分支实现（QKV tap、因果 KernelMatrix、低维累加循环、FFN 融合、M/F 协议）语义正确：出口全零时与未修改 Base 输出一致；两步梯度启动后 Q/K/V 压缩与 Rq/Ro 获得非零梯度且主干参数不更新；KernelMatrix 与 KernelSlot 的输出及 Q/K/V 梯度一致；Cg=PWg、Cu=PWu 约束下 C_dual 与 F_pre 的输出及对同一 P 的梯度一致；改动后缀不改变前缀输出；一次 prefill、分块 prefill 与逐 token decode 一致；padding 不写不读、reset/batch 重排不串状态；同 token 只写一次且 S 不改变写次数；M 线 support 主干 KV 不进入 query forward。(scope: Qwen/Qwen3-0.6B-Base（锁定 revision）；FP32 小张量参考用严格阈值；GPU BF16 容差由同一 Base 的重复测量预先锁定，看到失败后不得放宽；检查围绕 plan §6.3 列出的风险，不要求重跑无关实验)

- H-0002@1 (challenged, paused) FFN gate/up 条件融合优于加法写回（V1 tokenwise proxy）: F 线（保留原 attention 上下文）、4 层 [6,13,20,27]、r32、无侧支记忆与循环的 tokenwise proxy Z=SiLU(UA) 下，把同一 Z 注入冻结 FFN 的 gate/up（C_gate/C_up/C_dual）比加法写

回（F_out/F_pre）在非天花板 dev 切片上取得更好的任务准确率与更低 NLL；且该优势不能仅由线性退化组（Z=UA + C_dual，即共享 A 的 gate/up LoRA）或参数相近的 LoRA（4 层 all-7 rank10；28 层 all-7 rank2）解释。(scope: F 线；真实任务 train/dev（PIQA+BoolQ 官方 train 切分）或预先锁定的非天花板切片；每组 ≤2M 非 padding 输入 tokens、seed 0 单 seed 筛选（screening）；接口间不等参，只筛位置与信号，不据此宣称参数效率；无记忆 proxy 的负结果不否定有记忆 concat（V2 保留 C_dual 交互探针）)

- H-0003@1 (challenged, retired) M 线单跳：learned memory 读取产生可用信号（V2）: M 线（问答阶段丢弃 support 主干 KV，仅保留声明的低维 memory）、单层 0-based index14、r128、S=1、T1 单跳（E=16 事实、4 关系、visible_fact_delimiter 写入、all_valid_question 读取）、仅答案 CE 从 Base 训练时，2×2×2 析因（特征 direct_hidden/qknorm_pre_rope × 读算子 KernelMatrix/SoftmaxSlot × 融合 F*/C*）中至少一组在新 dev 上正常 memory 的 EM 同时比 zero 与 shuffled memory 高至少 10 pp 且 NLL 更低，并有可用正对照与非退化读出。(scope: study_id concat_loop_v2 的 V2 数据（独立 phase/seed 命名空间，不复用旧验证集）；每组 ≤4M 输入 tokens、单 seed 筛选；门槛须在 V2 release 前锁定；是否继续有限循环探索的研究门槛，不是统计确认，也不替代旧 writer 的 80% gate)

- H-0004@1 (inconclusive, paused) 读算子选择性：SoftmaxSlot 读取比 KernelMatrix 更集中且更可用: 在相同正核 q/k 特征、相同写读位置与训练条件下，SoftmaxSlot（s0=sqrt(dk)、tau=1、g=1）比 KernelMatrix 产生更高的目标 key 权重、更少的有效读取项数、更低的归一化读误差与更高的单跳 EM；若 SoftmaxSlot 有效而 KernelMatrix 无效，瓶颈在读权重的选择性与混合，而非 memory 中的信息缺失。(scope: V2 析因内的读算子主效应与交互；单 seed 筛选；差异只称读算子差异，不归因于“保存 KV 消除了矩阵压缩”；KernelSlot 只作恒等检查，不作训练 arm)

- H-0005@1 (untested, paused) 固定记忆下演化 query 的多次读取有独立收益（V3→V4）: 固定 memory（同一 token 的各步共享 M、c）下，累加循环 S=2/4 的 hops2–3 EM 优于独立训练的 S1，且该收益不被固定 query 精确幅度对照（Z=S·eta·a·Ro）、固定总步长、同源 tokenwise 核、MAC/延迟匹配的更宽 S1、同核锚定更新与完整状态读出解释。(scope: M 线，沿用 V2 选定的特征/融合/rank/scope；T2 可组合图任务（训练与 dev 仅 hops1–3，hops4/6/8 封存外推）；V3 每组 ≤5M 单 seed 筛选；确认（V4）需预注册：hops2–3 平均 EM 差 $\geq +3$ pp、配对区间下界 >0、3 个新 seeds 方向均正、hop1 差不低于 -1 pp)

- H-0006@1 (untested, candidate) 保留原 attention 的实用适配价值（V5）: F 线（4 层 r32，保留原 attention 上下文）的 PIQA+BoolQ 适配中，C* 多读相对参数匹配的强 LoRA（全 28 层 all-7）宏平均准确率提升至少 +1 pp 且每任务下降不超过 1 pp；并且侧支历史相对同参 tokenwise T0 与 zero/shuffle/permute 干预有可测贡献；同时报告延迟与状态成本。(scope: 官方 validation 封存为确认集，配置与分析锁定后只开一次；3 seeds × ≤12M、最多 3 epochs；95% 配对区间 + Holm 校正；WikiText-103 保留 split 的 NLL 作能力保持；参数、状态与计算不同时等量，不宣称同时公平)

- H-0007@1 (untested, paused) 冻结 S-RE：已学 K/V 换用槽位读算子后更可用（旧资产诊断）: 冻结旧 S-RE（DPFP 实际 key 768 / value 128、原读位置、memory-only）的 writer 与低维 reader，仅把原生 ridge read 在所有正常读取位置同时替换为 SoftSlot 或 HardSlot（温度/增益只在独立 calibration 上按 NLL 选择），旧 1,000 个验证 episode 的 EM 高于原生读取（旧参考 723/1,000，NLL 1.642）。(scope: 需要旧 S-RE 推理资产（writer.pt 与 looplora.llm.sq2.load_inference）；只回答既有 learned K/V 在另一读算子下能否更有效使用，不证明训练后极限；复现旧验证只证明兼容性；冻结诊断上限 1 node·h)

2 Related work

目前的记录中还没有核查过的文献条目，因此这里不对相关工作作任何断言。与本工作最直接相关的是同一计划此前的 LoopLora 实现（UnbearableFate/LoopLora @ a255e81）。本研究只把它作为只读的数据重叠核对来源：E-0005 的新数据与其 registry 重叠为 0。本研究没有复现它的结果；冻结 S-RE 诊断因缺少推理资产而暂停（D-0001）。V1 中的 LoRA 只作为参数相近的对照。系统的文献核查留待后续阶段。

- none recorded

3 Method

所有实验都使用 Qwen/Qwen3-0.6B-Base（锁定 revision da87bfb6），BF16 冻结主干、FP32 新参数、eager attention，硬件为 GH200 MIG 2g.24gb；所有工作负载都在 PBS 作业中运行。E-0001 与 E-0004 是不训练的实现检查，容差在检查前由同一 Base 的重复测量锁定（BF16 atol 0.50002，FP32 2e-5），失败后不放宽。E-0002 从 PIQA 与 BoolQ 的官方 train 中切出 train、calibration 与 dev（dev 每任务 1,000 条，官方 validation 封存不读），在三个代表结构上用 0.2M tokens 比较 1e-4 与 3e-4，按 calibration NLL 选学习率，三个家族都选中 3e-4。E-0003 用 seed 0 训练八组：proxy $Z = \mathrm{SiLU}(UA)$（4 层 [6,13,20,27]，r32）分别接 F_out、F_pre、C_gate、C_up、C_dual；线性退化组 $Z = UA$ 接 C_dual；以及 4 层 rank10 与 28 层 rank2 的 LoRA。每组在累计 2M 非 padding 输入 tokens 的那一步停止（13,077 步，batch 2），warmup 5% 后 cosine；主结果用最终 checkpoint，并要求重载后复评逐条一致。主指标是 dev 上 PIQA 与 BoolQ 准确率的宏平均。F*、C*、“有效”与“concat 胜出”的规则都在第一个正式 run 之前写进契约，按点估计判定；配对 bootstrap 区间（10,000 次，按任务分层）只作描述。E-0005 生成 V2 的单跳合成任务（每个 episode 16 条事实、4 种关系、单 token 答案），并在 dev 上评估 Base 的 M 线（只有问题）与 F 线（事实在上下文中）。E-0006 实现 M 线训练与诊断流水线并做冒烟。E-0007 在 calibration 切片上诊断 Base 在 F 线上失败的原因。V2 沿用 M 线协议：support 只写不读，问题阶段丢弃 support 的主干 KV，只保留单层（index 14）、r128、S=1 的低维记忆。E-0008 先训练事实在上下文中的 F 线正对照；E-0009 按家族在 0.2M tokens 上校准学习率（KernelMatrix 1e-2，SoftmaxSlot 3e-3）。E-0010 用 seed 0 训练十组，每组 6,000 步、4,000,470 tokens，按运行前锁定的门槛判定：正常记忆的 EM 同时比清零与打乱记忆高至少 10 pp、NLL 更低、目标权重至少 0.125。它还报告 oracle 与读取诊断，以及 10,000 次配对 bootstrap 区间（只作描述）。E-0012 只做测量，不设自动判定：提取与探针正对照（L0）；冻结第 14 层特征上的线性探针与以目标事实为标签的双线性寻址探针（L1）；训练后与随机初始化投影上的探针（L2）；计划 §8.3 的独立 clean-value 与 read-value ridge 探针（L3）；读取槽位、key 几何与融合幅度（L4）。E-0011 为 V3 生成了 T2 链终点数据，因 V3 暂停而没有使用。各实验的契约摘要如下。

E-0001 V0 实现与不变量检查

- question: 新侧分支实现能否通过 plan §6.3 的针对性检查（函数保持、两步梯度启动、冻结输入 Jacobian、同核恒等、约束融合恒等、因果、cache、reset/padding、所有权、read/write 计数、M 线信息访问、导出重载），从而可以进入训练？
- intervention: 在 src/ 实现 FeatureTap（qknorm_pre_rope 为主，另有 raw_qkv、direct_hidden；K/V 在 repeat_kv 之前压缩，不重复执行原投影）、LowDimProjection、MemoryBackend（kernel_matrix、kernel_slot_reference、soft_slot；M、c 用 FP32 累积，空记忆显式输出零）、LatentReader（累加与锚定更新，readout=increment 或 full，eta=0.25 与 g=1 固定）、Fusion（F_out、F_pre、F_attnres、C_gate、C_up、C_dual，出口零初始化）、Protocol（M 线 support write-only、query read-only、support 主干 KV 不进 query；F 线 inclusive/strict

因果写入）与 Accounting；原 attention 与原 FFN 每层每次前向只执行一次；用 pytest 在 Miyabi-G 作业中检查

- comparator: 未修改的 Qwen3-0.6B-Base 前向（同输入、同 dtype）；KernelSlot 参考读取；用同一 P 构造的 F_pre；一次 prefill 对分块 prefill 与逐 token decode；固定 query 时的精确幅度关系 Z=S·eta·a·Ro

- metric: plan §6.3 每项检查的 pass/fail；输出与梯度的最大绝对/相对误差对比预先锁定的容差（FP32 严格阈值；BF16 容差由同一 Base 的重复测量锁定，并在检查运行前写入配置）；各参数组的梯度范数；主干参数更新量（应为 0）；write/read/原块调用计数

- scope: Qwen/Qwen3-0.6B-Base（锁定 model 与 tokenizer revision）；F 线 layers [6,13,20,27]、M 线 layer [14]；r ∈ {32,128}；S ∈ {1,2,4}；FP32 小张量参考与真实模型（BF16 主干 + FP32 分支）两种设置；shared→untied 复制恒等推迟到 V3 之前

- contract: E-0001@1, sealed as DOC-0003@1 (research/experiments/E-0001.md)

E-0002 V1 任务切片与家族学习率校准

- question: PIQA+BoolQ（从官方 train 切出 train/dev/calibration，纯文本固定模板）在 4 层 r32 设置下是否处于可学习且未饱和的区间？LoRA、tokenwise 加法、tokenwise concat 三个家族在 calibration 上各取哪个学习率？

- intervention: Base 只评估不训练；三个家族各两个学习率 {1e-4, 3e-4}，每个 run 0.2M 非 padding 输入 tokens，seed 0，相同数据顺序：LoRA（4 层 [6,13,20,27] all-7 rank10）、tokenwise 加法（proxy Z=SiLU(UA) + F_out）、tokenwise concat（proxy Z=SiLU(UA) + C_dual）

- comparator: Base；同一家族的两个学习率之间

- metric: calibration NLL（选择依据，运行前固定）与准确率；Base 与 LoRA 在 dev 上的 PIQA、BoolQ 准确率（地板/天花板检查）；实测吞吐（tokens/s）、每步时间、加载/评估/保存时间与 node·h

- scope: F 线 context-retained；max length 512；AdamW betas (0.9, 0.999)、eps 1e-8、矩阵 weight decay 0.01；warmup 占实际优化步数 5% 后 cosine；global-norm clip 1.0 并记录 pre-clip 范数；dropout 0；PAD 与 EOS 同 ID 时按实际 padding 位置屏蔽；模板、标签 token、候选打分与多 token 标签归一化规则在本实验内固定并写入 release；官方 validation 不开封

- contract: E-0002@1, sealed as DOC-0019@1 (research/experiments/E-0002.md)

E-0003 V1 FFN 融合接口筛选（八组）

- question: tokenwise proxy 下，哪种接口能把低维状态转成任务收益：加法写回（F_out/F_pre）还是 FFN gate/up 条件融合（C_gate/C_up/C_dual）？这种优势是否超出线性退化组与参数相近的 LoRA？

- intervention: V1-01 至 V1-05：proxy Z=SiLU(UA)（4 层 [6,13,20,27]，r32）分别接 F_out、F_pre、C_gate、C_up、C_dual；V1-06：线性 Z=UA + C_dual

- comparator: V1-07：同 4 层 all-7 LoRA rank10（约 0.92M 参数）；V1-08：全 28 层 all-7 LoRA rank2（约 1.26M 参数）；V1-06 线性退化组；Base 只评估

- metric: dev 上 PIQA、BoolQ 准确率及其宏平均（主指标）与 dev NLL；stored、trainable 与实际有梯度的参数数；实测吞吐、wall time 与 node·h；所有 arm 在相同 dev 样本上评估
- scope: F 线；沿用 V1 切片与学习率校准锁定的模板、切分、各家族学习率与优化配方；seed 0 单 seed 筛选（screening），不作确认；max length 512；checkpoint 选择规则在运行前固定且所有 arm 相同；接口间不等参，不宣称参数效率；官方 validation 不开封
- contract: E-0003@2, sealed as DOC-0040@2 (research/experiments/E-0003.md)

E-0004 记忆路径就绪检查（RV-0002 跟进）

- question: V2 用到的记忆路径在分块缓存与支持写入容量两处是否正确：FP32 下 chunk3 的记忆状态是否与整段前向一致；支持写入容量保护是否对所有 backend 一致生效？
- intervention: 按 RV-0002 两条 low 跟进：为 chunk3 比较 cache.memory[14] 的 M、c（slot backend 还有 slots）与整段前向的 FP32 状态，并记录 cache.chunk3.state.max_abs_err；在 visible_fact_delimiter 的 support 阶段，对所有 backend（含 kernel_matrix）执行 counts <= slot_capacity
- comparator: 整段前向的记忆状态；slot backend 已有的容量保护行为
- metric: 检查通过数；chunk3 状态 max_abs_err（FP32 atol/rtol 2e-5，来自已锁定的 tolerances.json）；chunk1 状态误差只作诊断；kernel_matrix 在 slot_capacity=1 时按预期报错
- scope: 单层 index14、r128（V2 默认）与 r32；KernelMatrix、KernelSlot、SoftmaxSlot；GH200 MIG 2g.24gb；容差不改
- contract: E-0004@1, sealed as DOC-0047@1 (research/experiments/E-0004.md)

E-0005 M 线合成任务生成器与 Base 上下限（concat_loop_v2）

- question: 新的 concat_loop_v2 合成关系任务能否按 plan §8.1、§13.1–13.2 的协议确定性生成、与旧 LoopLora 数据不重叠，并且 Base 在 M 线（只有问题）下接近先验基线、在 F 线（支持事实在上下文中）下明显高于先验基线，从而适合 V2 的单跳记忆读取研究？
- intervention: 实现生成器、episode registry 与切分；生成 V2 的 T1 单跳任务（E=16 个支持事实、4 种关系、固定公开渲染模板、单 token 答案）：train、calibration 2,048、dev 1,000，以及只登记身份与哈希的封存 confirmation 2,000；多跳（训练/dev hops 1–3，封存外推 hops 4/6/8）只实现接口与身份；Base 只在 dev 上评估 M 线与 F 线
- comparator: 旧 LoopLora registry（UnbearableFate/LoopLora @ a255e81，只读）；先验基线（按 train 答案频率的最佳常数答案）；M 线 Base 对 F 线 Base
- metric: 两次生成哈希一致；各 split 的 episode 身份互不相交；与旧 registry 重叠为 0；答案都是单 token；各轴统计（关系、事实数、干扰项、文本长度、名字池与 tokenization 长度、支持顺序）；每条事实恰好一个写入分隔符位置；Base dev 的完整词表 EM 与答案 NLL（M 线、F 线）
- scope: study_id=concat_loop_v2，phase V2；seed 由 derive_seed(20261029, "concat_loop_v2", phase, purpose) 派生，derive_seed 与 T-0002 的定义相同；write_scope=visible_fact_delimiter，read_scope=all_valid_question；不训练；封存切片不做任何模型评估；GH200 MIG 2g.24gb
- contract: E-0005@1, sealed as DOC-0048@1 (research/experiments/E-0005.md)

E-0006 V2 M 线训练与诊断流水线（实现与冒烟）

- question: 按 plan §8.1 的 M 线协议（support write-only、query read-only、单层 index14、r128、S=1）实现的训练与评估流水线，以及 §8.3、§8.4 规定的诊断，是否在小规模冒烟中行为正确（初始化时与 Base 相同、梯度按预期流动、因果对照与 oracle 开关按定义改变读出）?
- intervention: 实现 M 线训练/评估入口：support 阶段只写、问题阶段从同一只读快照读取、按 episode owner 索引、单 token 答案 CE、完整词表 EM 与 NLL；§8.3 诊断（寻址 top-1、margin、目标权重、熵、n_eff，读误差，输出范数，数值统计）；zero / shuffled / permuted memory 因果对照；§8.4 oracle 开关（真实 key 替换 query、直接提供目标 value、zero memory、其他 episode 的 memory，只作离线评估）
- comparator: Base（出口零初始化时应与之完全相同）；正对照小任务上 normal memory 对 zero memory
- metric: 检查通过数；初始化时与 Base 的 logits 差；各控制与 oracle 开关的读出变化；冒烟步时、显存与吞吐（为记忆家族校准与 V2 估算成本）
- scope: 特征 direct_hidden 与 qknorm_pre_rope；读算子 KernelMatrix 与 SoftmaxSlot（g=1，s0=sqrt(dk)，tau=1）；融合 F_out 与 C_dual；write_scope=visible_fact_delimiter，read_scope=all_val GH200 MIG 2g.24gb；只做冒烟，不做正式训练
- contract: E-0006@1, sealed as DOC-0053@1 (research/experiments/E-0006.md)

E-0007 F 线 Base 难度诊断（concat_loop_v2）

- question: E-0005 中 Base 的 F 线 EM 接近先验（dev 1.3%，先验 1.5%）；协调者只读核查已有预测，dev 上 F 线 argmax 有 94.4% 是问题主语，M 线为 76.7%。Base 是在上下文中找到了正确实体、只是被复制问题主语的倾向压过，还是在 E=16 时根本没有找到？失败是否来自提示格式或评估实现？
- intervention: 在 calibration 切片上沿用 E-0005 的 Base 提示与评分路径，逐条输出完整词表 top-5、正确答案的秩与概率、问题主语的概率与 top-1 类别；计算限定候选准确率；F 线另做事实数梯度 E ∈ {1,2,4,8,16}：保留被问事实，其余事实按派生 seed 确定性抽取并保持原相对顺序，用生成器原模板重新渲染
- comparator: E-0005 的 Base dev 结果（M 1.1%、F 1.3%、先验 1.5%）；同一 calibration 上的 M 线；按候选数计算的随机基线；E=1 作为最容易的正对照
- metric: 完整词表 EM 与答案 NLL；正确答案的秩与概率；问题主语复制率；top-1 类别分布；限定候选准确率：(a) 支持中出现的全部不同名字，(b) 同一集合去掉问题主语，(c) 与被问事实同关系的事实的不同尾实体；按 E、被问事实位置、关系、support_order 与 answer_frequency_stratum 分层
- scope: 只用 Base（Qwen3-0.6B-Base @ da87bfb608c14b7cf20ba1ce41287e8de496c0cd），不训练；只用 calibration 切片（runs/ATT-0039-1/data/calibration.jsonl，sha256 c08753365165216b5a9f4c3d629 dev 不再评估，confirmation 不读取；E 变体只是诊断视图，不是新切分；不改 E-0005 的协议、模板与数据；diagnostic 证据，不支持也不挑战 H-0003
- contract: E-0007@2, sealed as DOC-0063@2 (research/experiments/E-0007.md)

E-0008 V2 F 线训练后正对照（事实在上下文中）

- question: 在与 V2 M 线相同的数据、样本顺序与 4M 输入 tokens 预算下，事实放在上下文中、只用答案 CE 训练的标准 LoRA，能否在 E=16 的单跳任务上明显超过先验？即任务本身在该规模与预算下是否可学，从而 M 线结果可以解读。
- intervention: LoRA（4 层 [6,13,20,27]，all-7 模块，rank 10，与 V1-07 相同布局）接在冻结 Base 上；F 线提示 = support_text + question_text（与 E-0005/E-0007 的 F 线评估相同），只在答案位置计算 CE；学习率在 {1e-4, 3e-4, 1e-3} 上各跑 0.2M tokens，按 calibration 答案 NLL 选择；选中后跑完整的 4M tokens
- comparator: E-0005/E-0007 的未训练 Base F 线（calibration EM 1.42%，E=16）；按 train 答案频率的最佳常数答案（先验基线）
- metric: calibration 与 dev 上的完整词表 EM 与答案 NLL；答案不等于问题主语的子集上的 EM；问题主语复制率；学习曲线；吞吐、wall time 与 node·h
- scope: concat_loop_v2 V2 数据（E-0005，train 16,384 / calibration 2,048 / dev 1,000）；seed 0；F 线结果单独登记，不替代 M 线；confirmation 不读取；只作为 V2 的正对照与第二步的放行门槛
- contract: E-0008@1, sealed as DOC-0086@1 (research/experiments/E-0008.md)

E-0009 V2 记忆家族学习率校准与正式 M 线训练程序

- question: 正式 M 线训练程序（4M token 日程、对照评估、checkpoint 重载）能否正确运行？在 V2 共同配方下，KernelMatrix 与 SoftmaxSlot 两个读算子各应使用哪个学习率？
- intervention: 实现正式 M 线训练程序（复用 E-0006 的 collate、train_forward、evaluate_batch 与对照，加入 C_up）；代表配置 qknorm_pre_rope + F_out（层 14，r128，S=1）分别接 kernel_matrix 与 soft_slot，学习率 {3e-4, 1e-3, 3e-3} 各跑 0.2M tokens
- comparator: 同一 run 中 zero memory 与 shuffled memory 的 calibration 答案 NLL（诊断）；E-0006 的冒烟结果
- metric: calibration 答案 NLL（normal memory，选择依据）；zero 与 shuffled 的 NLL 与 EM（诊断）；梯度范数与裁剪比例；吞吐、显存、wall time 与 node·h
- scope: concat_loop_v2 V2 数据；seed 0；V2 共同训练配方（E-0008 Design 段）；校准只选学习率，不作 H-0003 的证据；选中的学习率用于 E-0010 中同一读算子的全部 arm（包括 C_dual 探针）
- contract: E-0009@1, sealed as DOC-0087@1 (research/experiments/E-0009.md)

E-0010 V2 M 线单跳记忆读取：2×2×2 析因与 C_dual 探针

- question: M 线（问答阶段丢弃 support 主干 KV，只保留低维 memory）下，哪种特征 × 读算子 × 融合组合能让 learned memory 读取产生可用信号（normal memory 明显优于 zero 与 shuffled）? SoftmaxSlot 是否比 KernelMatrix 读得更集中、更可用？
- intervention: 十组：特征 direct_hidden / qknorm_pre_rope × 读算子 kernel_matrix / soft_slot × 融合 F*=F_out / C*=C_up（V2-01 至 V2-08），以及 qknorm_pre_rope × {kernel_matrix, soft_slot} × C_dual 交互探针（V2-09、V2-10）；层 14，r128，S=1，只用答案 CE，每组 4M 输入 tokens

- comparator: 同一 checkpoint 的 zero memory、shuffled memory 与 permuted association；E-0008 的 F 线训练后正对照；key/value oracle（诊断）；先验基线
- metric: dev 上 normal/zero/shuffled/permuted 的完整词表 EM 与答案 NLL（主指标为 normal 对 zero 与 shuffled 的 EM 差与 NLL 差）；读取诊断（目标权重、n_eff、熵、读误差）；答案 $\neq$ 问题主语子集上的 EM；学习曲线；参数数、吞吐、wall time 与 node·h
- scope: concat_loop_v2 V2 数据（E-0005），seed 0 单 seed 筛选，不是确认；V2 共同训练配方（E-0008 Design 段）；各读算子的学习率来自 E-0009；判定规则在任何 V2 run 之前锁定；M 线结果与 F 线正对照分开登记；不替代旧 writer 的 80% gate；confirmation 不读取
- contract: E-0010@2, sealed as DOC-0083@2 (research/experiments/E-0010.md)

E-0011 V3 T2 链终点任务生成器、登记与难度检查

- question: 能否按 plan §9.1、§13.1–13.2 确定性地生成 V3 的 T2 链终点任务（训练/dev 只含 hops 1–3，封存 hops 4/6/8 与未见名字），与 V2 数据和旧 registry 不重叠，结构捷径留有余量，并且 M 线 Base 接近先验、F 线 LoRA 小 pilot 能学会 hops 1–3？
- intervention: 新模块 src/v3_data.py（不改 src/v2_data.py）：phase 参数化的 T2 链终点生成器、episode registry 与切分；生成 phase V3 的 train/calibration/dev，并登记封存切片的身份与哈希；计算无模型的结构捷径基线；Base 在 dev 上评估 M 线与 F 线；F 线 LoRA 小 pilot（复用合并后的 src/v2_fline.py，0.2M tokens）
- comparator: 先验基线（train 最佳常数答案）；无模型结构捷径基线（随机 r 终点猜测、忽略关系类型走到底、最长 r 链终点、支持中最后出现的 r 终点）；V2 T1 数据（ATT-0039-1）与旧 LoopLora registry 的内容哈希
- metric: 两次生成哈希一致；身份与语义内容在切片内外互不重复，与 V2 和旧 registry 重叠为 0；单 token 答案与答案边界检查；各轴统计（hop、链长构成、r 链数、干扰事实位置、支持顺序、名字频率层、tokens）；每个 hop 的结构捷径 EM；M 线 Base 与 F 线 Base 的 dev 完整词表 EM/NLL（分 hop）；LoRA pilot 的 calibration EM/NLL（分 hop）；node·h
- scope: study_id=concat_loop_v2，phase V3；derive_seed(20261029, "concat_loop_v2", "V3", purpose)；write_scope=visible_fact_delimiter；E=16；只用 V2 的 seen 名字池（unseen 只用于封存切片）；封存切片不做任何模型评估；不训练 M 线；GH200 MIG 2g.24gb
- contract: E-0011@2, sealed as DOC-0097@2 (research/experiments/E-0011.md)

E-0012 V2 记忆失败层次定位（表示探针与寻址几何）

- question: E-0010 十组都没有读出 episode 记忆（value oracle 也无效）。在 M 线的各层次上，答案相关信息的线性可解码性在哪里下降：冻结第 14 层特征（写入位置的尾实体、问题特征对目标事实的寻址）、训练后的 key/value/query 投影、学到的目标 value（clean-value probe）、实际读取向量（read-value probe），以及读取的槽位与融合幅度？测量结果供协调者选择 plan §8.4 允许的唯一探针，或以预期价值有限为由停止扩大记忆路线。
- intervention: 不训练 M 线：L0 提取与探针正对照；L1 冻结 Base 第 14 层特征（direct_hidden；qknorm_pre_rope 的 q_norm/k_norm 与 v_proj 输出）上的线性探针与双线性寻址探针（秩 128/256/512/满秩）；L2 E-0010 十个 checkpoint 与随机初始化投影的 key/value/query 探针；L3 plan §8.3 的独立 clean-value 与 read-value 探针（src/v2_probe.py 路径）；L4 读取槽位分布、key 几何与融合输出幅度

- comparator: 随机水平（名字 1/92，关系 1/4，寻址 1/16）与按 train 频率的常数基线；L0 正对照；随机初始化投影对训练后投影；E-0010 的 oracle 与读取诊断
- metric: calibration 上各探针的 top-1 准确率（按层次、特征集、位置、目标与秩，附正确数与样本数）；clean_to_clean、read_to_read、clean_to_read；读取 argmax 槽位分布；episode 内 key 两两余弦与目标 margin；问题位置融合输出范数与第 14 层残差范数之比；node·h
- scope: V2 T1 数据（ATT-0039-1）train、calibration 与 dev，seed 0；不读 confirmation；不训练 M 线，不改 E-0010 结论；只作诊断测量
- contract: E-0012@1, sealed as DOC-0117@1 (research/experiments/E-0012.md)

4 Results

下列表格给出各实验可用 attempt 的描述性均值，不是统计检验。E-0001 与 E-0004 是 diagnostic 证据：V0 修正后 85/85 项、E-0004 130/130 项检查通过。E-0003 是单 seed 的 screening 证据：Base 的 dev 宏平均为 70.15%，各训练组在 77.05%–79.20% 之间。大部分提升来自 BoolQ（Base 68.0%，训练组 80.8%–83.6%）；PIQA 只从 72.3% 升到 73.0%–74.9%。按预先固定的规则，F*=F_out（77.80%），C*=C_up（77.70%）。最好的 concat 组 C_dual（78.15%）只比 F* 高 0.35 pp（描述区间 [−0.60, 1.35]），记为 concat 未胜。C* 比 4 层与 28 层 LoRA 分别低 1.50 与 1.35 pp；C_dual 对线性退化组只高 0.15 pp [−0.70, 1.00]。E-0005 与 E-0007 是 diagnostic 证据。Base 在 dev 上的 M 线与 F 线 EM 分别为 1.1% 与 1.3%，先验基线为 1.5%。在 calibration 上，上下文事实数 E 为 1、2、4、8、16 时，F 线 EM 依次为 67.0%、6.9%、2.2%、1.4%、1.4%；$E \geq 2$ 时，91%–96% 的回答是问题主语；E=16 时，在支持中的名字（去掉主语）里选对的比例为 14.2%，随机为 3.7%。E-0007 有一项实现检查按原规则未通过：BF16 下 batch 1 与 batch 8 的逐条 NLL 平均绝对差为 0.088，门槛为 0.05。FP32 对照把该差异降到 1e-5 量级，因此判为 BF16 数值敏感。

V2 的结果如下。E-0008 的 F 线正对照通过：事实在上下文中训练后，calibration EM 为 99.95%（训练前 1.90%）。E-0010 十组的 dev 正常 EM 为 0.8%–2.1%，清零记忆为 1.1%，打乱记忆为 1.1%–1.9%；正常 EM 对两者中较大者的差最大为 +0.6 pp。所有组的平均目标权重都低于 0.125，也低于均匀的 1/16。正常与打乱记忆的 NLL 几乎相同（4.42–4.61），清零记忆为 9.25。把读取换成目标的已学 value（value oracle）后，EM 为 1.1%–2.3%。按锁定规则没有合格组，H-0003 记为 challenged（screening）；H-0004 的读算子比较中，两种算子都只在 0/4 格占优，记为 inconclusive。E-0012 的六项正对照为 96.6%–100%。在分隔符位置，冻结特征解码尾实体的准确率为 99.6%（direct_hidden）与 92.9%（qknorm_pre_rope）。以目标事实为标签的双线性寻址探针在最后一个问题位置、秩 128–1024 下的 16 选 1 top-1 为 82.2%–90.2%，秩增大没有一致的提升。训练后，每组在全部 1,000 个 dev episode 的答案位置都把第一个写入的槽位排在最前。独立 ridge 探针下，读出向量的答案准确率为 1.8%–2.3%，已学目标 value 为 16.6%–23.0%。逐层 softmax 探针中，训练后 value 投影解码尾实体的准确率（17.0%–37.1%）不高于随机初始化投影（32.2%–38.5%）。融合增量与第 14 层残差范数之比的组均值为 18.7–608（V2-09 为 1.1e6）。

Table 1: E-0001 V0 实现与不变量检查: descriptive means over usable attempts

arm	seeds	attempts	bf16_atol	check.src/test_core.py::test_kernel_matrix_eq
base_calibration	173	1 ended/success	0.5 (n=1)	-
hardware_probe	0	1 ended/failed, 1 ended/success	-	-
invariant_checks	42	2 ended/failed, 2 ended/success	-	1.176e-08 (n=2)

Per-attempt records: rl run show E-0001.

Table 2: E-0002 V1 任务切片与家族学习率校准: descriptive means over usable attempts

arm	seeds	attempts	C_dual.calibration.boolq.accuracy	C_dual.calibra
base-eval	0	1 ended/cancelled, 1 ended/success	-	-
cal-cdual-1e-4	0	1 ended/cancelled, 1 ended/success	-	-
cal-cdual-3e-4	0	1 ended/cancelled, 1 ended/success	-	-
cal-fout-1e-4	0	1 ended/cancelled, 1 ended/success	-	-
cal-fout-3e-4	0	1 ended/cancelled, 1 ended/success	-	-
cal-lora-1e-4	0	1 ended/cancelled, 1 ended/success	-	-
cal-lora-3e-4	0	1 ended/cancelled, 1 ended/success	-	-
release-schedule-estimate	0	1 ended/success	-	-
selection-boundary-audit	0	1 ended/success	-	-
selection-release-audit	0	1 ended/failed	-	-
smoke	0	1 ended/success	0.6562 (n=1)	32 (n=1)
smoke-corrected	0	1 ended/success	0.75 (n=1)	32 (n=1)
smoke-reloadfix	0	1 ended/success	0.75 (n=1)	32 (n=1)
smoke-reviewfix	0	1 ended/success	0.75 (n=1)	32 (n=1)

Per-attempt records: rl run show E-0002.

Table 3: E-0003 V1 FFN 融合接口筛选（八组）: descriptive means over usable attempts

arm	seeds	attempts	C_dual.dev.boolq.accuracy	C_dual.dev.boolq.example
Base	0	1 ended/success	-	-
V1-01	0	1 ended/success	-	-
V1-02	0	1 ended/success	-	-
V1-03	0	1 ended/success	-	-
V1-04	0	1 ended/success	-	-
V1-05	0	1 ended/success	-	-
V1-06	0	1 ended/success	-	-
V1-07	0	1 ended/success	-	-
V1-08	0	1 ended/success	-	-
analysis-regression	0	1 ended/success	-	-
base-dev-comparison	0	1 ended/success	-	-
formal-analysis	0	1 ended/success	-	-
smoke-eight-arms	0	1 ended/failed, 1 ended/success	0.625 (n=1)	32 (n=1)

Per-attempt records: rl run show E-0003.

Table 4: E-0004 记忆路径就绪检查（RV-0002 跟进）: descriptive means over usable attempts

arm	seeds	attempts	check.src/test_core.py::test_kernel_matrix_equals_slot_reference_
readiness	42	1 ended/success	1.176e-08 (n=1)

Per-attempt records: rl run show E-0004.

Table 5: E-0005 M 线合成任务生成器与 Base 上下限 (concat_loop_v2) : descriptive means over usable attempts

arm	seeds	attempts	F.elapsed_s	F.em	F.examples	F.nll	M
base-dev-MF	-	1 ended/success	4.633 (n=1)	0.013 (n=1)	1000 (n=1)	5.706 (n=1)	4.
base-dev-smoke	-	1 ended/success	0.1641 (n=1)	0 (n=1)	32 (n=1)	5.918 (n=1)	0.
data-generation	-	1 ended/success	-	-	-	-	-
data-smoke	-	1 ended/success	-	-	-	-	-
legacy-inventory	-	1 ended/success	-	-	-	-	-

Per-attempt records: rl run show E-0005.

Table 6: E-0006 V2 M 线训练与诊断流水线 (实现与冒烟) : descriptive means over usable attempts

arm	seeds	attempts	arms_completed	check.src/test_core.py::te
eight-arm-E0005-integration	2100042	1 ended/success	-	-
eight-arm-copy32	2100042	1 ended/success	8 (n=1)	-
eight-arm-timing	2100042	1 ended/success	8 (n=1)	-
invariant-checks	2100042	1 ended/success	-	1.176e-08 (n=1)
shuffled-diagnostic-regression	2100042	1 ended/failed, 1 ended/success	-	-

Per-attempt records: rl run show E-0006.

Table 7: E-0007 F 线 Base 难度诊断 (concat_loop_v2) : descriptive means over usable attempts

arm	seeds	attempts	F.E1.em	F.E1.nll	F.
diagnostic-smoke32	20261029	1 ended/failed	-	-	-
diagnostic-smoke32-fixed	20261029	1 ended/success	0.625 (n=1)	1.519 (n=1)	0.
fp32-calibration256-and-saved-bf16-audit	173	1 ended/success	-	-	-
full-calibration2048	20261029	1 ended/success	0.6699 (n=1)	1.422 (n=1)	0.
postmerge-regression	20261029	1 ended/success	-	-	-
scoring-difference-audit	20261029	1 ended/success	-	-	-

Per-attempt records: rl run show E-0007.

Table 8: E-0008 V2 F 线训练后正对照 (事实在上下文中) : descriptive means over usable attempts

arm	seeds	attempts	calibration.all.em	calibration.all.examples	calibrati
fline-4m	0	1 ended/success	0.9995 (n=1)	2048 (n=1)	0.01557 (
fline-cal-0.0001	0	1 ended/success	0.9941 (n=1)	2048 (n=1)	0.02509 (
fline-cal-0.0003	0	1 ended/success	0.998 (n=1)	2048 (n=1)	0.009097
fline-cal-0.001	0	1 ended/success	0.9995 (n=1)	2048 (n=1)	0.001602
fline-cal-0.003	0	1 ended/success	0.998 (n=1)	2048 (n=1)	0.0107 (n
fline-output-audit	0	1 ended/success	-	-	-
fline-review-regression	0	1 ended/success	0.0625 (n=1)	32 (n=1)	3.515 (n=
fline-smoke	0	1 ended/failed, 1 ended/success	0.0625 (n=1)	32 (n=1)	3.515 (n=

Per-attempt records: rl run show E-0008.

Table 9: E-0009 V2 记忆家族学习率校准与正式 M 线训练程序: descriptive means over usable attempts

arm	seeds	attempts	200000.budget	200000.steps	200000.t
final-grid-analysis	0	1 ended/success	-	-	-
formal-path-checks	0	1 ended/success	-	-	-
initial-grid-analysis	0	1 ended/success	-	-	-
kernel_matrix-lr0.0003	0	1 ended/success	-	-	-
kernel_matrix-lr0.001	0	1 ended/success	-	-	-
kernel_matrix-lr0.003	0	1 ended/success	-	-	-
kernel_matrix-lr0.01-extension	0	1 ended/success	-	-	-
reporting-max-regression-cost-report	0	1 ended/success	-	-	-
schedule-preflight	0	1 ended/success	2e+05 (n=1)	300 (n=1)	2.002e+0
soft_slot-lr0.0003	0	1 ended/success	-	-	-
soft_slot-lr0.001	0	1 ended/success	-	-	-
soft_slot-lr0.003	0	1 ended/success	-	-	-
soft_slot-lr0.01-extension	0	1 ended/success	-	-	-
twelve-arm-32step-smoke	0	1 ended/success	-	-	-

Per-attempt records: rl run show E-0009.

Table 10: E-0010 V2 M 线单跳记忆读取：2×2×2 析因与 C_dual 探针: descriptive means over usable attempts

arm	seeds	attempts	H3.eligible	H3.qualified	H4.kernel_wins	H4.soft_win
V2-01	0	1 ended/success	-	-	-	-
V2-02	0	1 ended/success	-	-	-	-
V2-03	0	1 ended/success	-	-	-	-
V2-04	0	1 ended/success	-	-	-	-
V2-05	0	1 ended/success	-	-	-	-
V2-06	0	1 ended/success	-	-	-	-
V2-07	0	1 ended/success	-	-	-	-
V2-08	0	1 ended/success	-	-	-	-
V2-09	0	1 ended/success	-	-	-	-
V2-10	0	1 ended/success	-	-	-	-
analysis-regression	0	1 ended/success	-	-	-	-
factorial-analysis	0	1 ended/success	0 (n=1)	0 (n=1)	0 (n=1)	0 (n=1)

Per-attempt records: rl run show E-0010.

Table 11: E-0011 V3 T2 链终点任务生成器、登记与难度检查: descriptive means over usable attempts

arm	seeds	attempts	F.all.em	F.all.examples	F.all.nll	F.all
F-LoRA-pilot	0	1 ended/success	-	-	-	-
base-dev	173	1 ended/success	0.009 (n=1)	1000 (n=1)	5.568 (n=1)	0.011
base-smoke	173	1 ended/success	0.03333 (n=1)	60 (n=1)	5.771 (n=1)	0.033
full-data	20261029	1 ended/success	-	-	-	-
generator-smoke	20261029	1 ended/success	-	-	-	-
generator-smoke-r2	20261029	1 ended/success	-	-	-	-
pilot-selector-smoke	0	1 ended/success	-	-	-	-

Per-attempt records: rl run show E-0011.

Table 12: E-0012 V2 记忆失败层次定位（表示探针与寻址几何）: descriptive means over usable attempts

arm	seeds	attempts	L3_s	address_s	dev_question_tokens	elap
L0-L4-artifact-audit	0	1 ended/success	-	-	7000 (n=1)	0.77
L0-L4-cost	0	1 ended/success	555.5 (n=1)	111.3 (n=1)	-	57.9
L0-L4-full	0	1 ended/success	-	-	-	945
L0-L4-smoke	0	1 ended/success	-	-	-	25.0

Per-attempt records: rl run show E-0012.

Evidence (level, validity):

- EV-0001 [diagnostic, valid] Allocated environment probe passed on mg1098 GH200 MIG 2g.24gb; driver615.71.09; outbound HTTPS available. No adapter correctness evidence yet.; scope: Execution environment only [code review: ATT-0001-2 not needed: Environment discovery only; no model or core operator execution]
- EV-0002 [diagnostic, valid] 54/60 checks passed; six failures all at suffix-causality comparison including two padded rows. Cache comparisons before assertion passed. Pinned eager masking uses finite minimum then softmax, so padded query outputs need Base diagnosis; no hypothesis conclusion.; scope: Implementation checks with pending padding-test diagnosis [code review: ATT-0003-1 not needed: Deterministic E0 invariant checks before code review; no training or costly run]
- EV-0003 [diagnostic, valid] 61/61 V0 tests passed on GH200 MIG 2g.24gb, fixed FP32 and Base-calibrated BF16 tolerances. Zero exits match Base; two-step grads and frozen backbone checks pass; kernel output/QKV gradient and constrained-fusion identities, causality on valid tokens, cache, padding/reset, ownership, counts, M-line access and export/reload pass. Base-only diagnostic confirms previous failures were padded query logits (13.443359375 vs valid-prefix 0). Code review RV-0001 pending; no training or mechanism claim. (supports H-0001@1) [code review: ATT-0004-1 not needed: Deterministic E0 checks and Base padding diagnosis; unchanged production code and tolerances]
- EV-0004 [diagnostic, valid] RV-0001 fixes: 79/85 pass; six new positive-control failures measure rank32 effects only .359-.781 and swapped support effect .383, below detection thresholds based on unchanged BF16 calibration. Strengthen only nonzero test exits to demonstrate

sensitivity before recheck; no production initialization change.; scope: Positive-control fixture adequacy [code review: ATT-0005-1 not needed: Deterministic checks of RV-0001 fixes before recheck; no training]

- EV-0005 [diagnostic, valid] 85/85 expanded V0 checks pass after RV-0001 fixes: real mutation controls rejected, delimiter support mask/capacity and stale-cache/missing-memory guards pass, masked/reset/empty kernel gradient identities pass, observed read accounting and configured key epsilon verified, actual fusion hook gradient identity passes. Nonzero fixture effects1.84375-6.890625 and swapped-owner effects3.53125-3.6484375 exceed unchanged BF16 thresholds. No training or mechanism claim; recheck pending. (supports H-0001@1) [code review: ATT-0006-1 not needed: Deterministic review-fix checks with stronger positive-control fixture; fixed production init and tolerances]

- EV-0006 [diagnostic, valid] V1 initial smoke:96 invariant tests passed and three20-step train/eval/reload paths passed;11586 train tokens. Training loss reduction subsequently superseded by coordinator MSG-0025; smoke establishes engineering compatibility/timing only, not LR or interface evidence. [code review: ATT-0007-1 not needed: deterministic regressions and 20-step three-family smoke before code review]

- EV-0007 [diagnostic, valid] Corrected equal-example V1 smoke:98 invariant tests passed including loss reduction and failed-LR continuation;three5-step training/evaluation/reload paths passed with exact reload equality.2943tokens,116s,.032222MIGh. Engineering diagnostic only; no LR selection or interface comparison. [code review: ATT-0008-1 not needed: corrected loss and divergence invariant regression plus5-step three-family smoke before review]

- EV-0008 [diagnostic, valid] RV0003fix smoke:104 invariant tests passed, including nongold NaN/Inf evaluation, retained failed-LR usage/continuation, Base nonfinite and finite reload mismatch errors, strict smoke acceptance.3families completed5steps and exact reload,2943tokens,114s,.031667 No calibration or interface claim. [code review: ATT-0009-1 not needed: RV-0003 failure-path fixes deterministic checks and5step smoke before recheck]

- EV-0009 [diagnostic, valid] Reload classification fix:106checks passed, including finite initial evaluation then actual NaN/Inf candidate guard during reload -> partial engineering error retaining usage without failed-LR flags. Existing initial-evaluation failure and finite reload mismatch behaviors pass.3families complete5steps/exact reload;2943tokens,119s,.033056MIGh=.008264nodeh. [code review: ATT-0010-1 not needed: RV-0004 reload error classification regression before coordinator-required new review round]

- EV-0010 [diagnostic, valid] base-eval complete finite calibration NLL 2.3312495373933837, calibration accuracies PIQA 0.72802734375 BoolQ 0.689453125; dev PIQA 0.723 BoolQ 0.68; exact reload passed. Base no training. Diagnostic LR/slice evidence only, not interface comparison. [code review: ATT-0011-2 covered by RV-0004 issues, medium open, approved by the coordinator, RV-0005 pass, RV-0006 pass]

- EV-0011 [diagnostic, valid] cal-lora-1e-4 complete finite calibration NLL 1.6253322842401003, calibration accuracies PIQA 0.7255859375 BoolQ 0.78955078125; dev PIQA 0.723 BoolQ 0.766; exact reload passed. 200280 input tokens,1103 steps,280 overshoot. Diagnostic LR/slice evidence only, not interface comparison. [code review: ATT-0012-2 covered by RV-0004 issues, medium open, approved by the coordinator, RV-0005 pass, RV-0006 pass]

- EV-0012 [diagnostic, valid] cal-lora-3e-4 complete finite calibration NLL 1.6021037854622986, calibration accuracies PIQA 0.728515625 BoolQ 0.80419921875; dev PIQA 0.733 BoolQ 0.78; exact reload passed. 200280 input tokens,1103 steps,280 overshoot. Diagnostic LR/slice evidence only, not interface comparison. [code review: ATT-0013-2 covered by RV-0004 issues, medium open, approved by the coordinator, RV-0005 pass, RV-0006 pass]

- EV-0013 [diagnostic, valid] cal-fout-1e-4 complete finite calibration NLL 1.6600830964101538, calibration accuracies PIQA 0.71923828125 BoolQ 0.7958984375; dev PIQA 0.723 BoolQ 0.777; exact reload passed. 200280 input tokens,1103 steps,280 overshoot. Diagnostic LR/slice evidence only, not interface comparison. [code review: ATT-0014-2 covered by RV-0004 issues, medium open, approved by the coordinator, RV-0005 pass, RV-0006 pass]

- EV-0014 [diagnostic, valid] cal-fout-3e-4 complete finite calibration NLL 1.6157966926032628, calibration accuracies PIQA 0.72119140625 BoolQ 0.79833984375; dev PIQA 0.725 BoolQ 0.78; exact reload passed. 200280 input tokens,1103 steps,280 overshoot. Diagnostic LR/slice evidence only, not interface comparison. [code review: ATT-0015-2 covered by RV-0004 issues, medium open, approved by the coordinator, RV-0005 pass, RV-0006 pass]

- EV-0015 [diagnostic, valid] cal-cdual-1e-4 complete finite calibration NLL 1.6392950516932028, calibration accuracies PIQA 0.72607421875 BoolQ 0.7919921875; dev PIQA 0.722 BoolQ 0.767; exact reload passed. 200280 input tokens,1103 steps,280 overshoot. Diagnostic LR/slice evidence only, not interface comparison. [code review: ATT-0016-2 covered by RV-0004 issues, medium open, approved by the coordinator, RV-0005 pass, RV-0006 pass]

- EV-0016 [diagnostic, valid] cal-cdual-3e-4 complete finite calibration NLL 1.6024291552247223, calibration accuracies PIQA 0.72705078125 BoolQ 0.798828125; dev PIQA 0.728 BoolQ 0.777; exact reload passed. 200280 input tokens,1103 steps,280 overshoot. Diagnostic LR/slice evidence only, not interface comparison. [code review: ATT-0017-2 covered by RV-0004 issues, medium open, approved by the coordinator, RV-0005 pass, RV-0006 pass]

- EV-0017 [diagnostic, valid] Deterministic E0002 audit selects3e-4 allfamilies; relative NLL gaps1.4499%,2.7408%,2.3006%; slice gate passes. E3 full train audit finds exactly1PIQA row id5528 candidates537/537tokens (.007654% of13065), BoolQ0/6379. Not in calibration prefix. E3 filter authorized MSG0058, production implementation deferred next task. No H0002 interface claim. [code review: ATT-0019-1 not needed: deterministic selection audit and identification of overlength E3 examples; unchanged data/encoding rules]

- EV-0018 [diagnostic, valid] Diagnostic-only post-shuffle/pre-interleave exclusion of PIQA5528 under MSG0058 yields E3 schedule13077steps,2000040tokens,26154examples,epoch index1; dev steps3270/6539/9808. Production filter not implemented; next task must implement/review and verify same schedule. Costs are family-based projections,28layer LoRA only conservative unmeasured bound. [code review: ATT-0020-1 not needed: cost/schedule estimate only under coordinator MSG0058; production filter and review deferred next task]

- EV-0019 [diagnostic, invalid] ATT-0021-1: 116 passed, one new audit fixture incorrectly expected success for 100 percent excluded train rows; actual guard correctly rejected >1 percent. No arm trained. Fixture repaired to exactly 1 percent without changing production guard. PBS21s, 0.005833333333333333 MIGh / 0.0014583333333333333 nodeh.; scope: Engineering regression fixture failure, not scientific evidence [code review: ATT-0021-1 not needed: Pre-review regression and twenty-step smoke for all eight arms]

- EV-0020 [diagnostic, valid] 126 tests passed; all eight arms match Base at initialization and reload exactly after20steps; parameter counts match contract, intermediate evaluation preserves training trajectory in regression, identical full 13077step/2000040token schedule with one excluded PIQA5528 row and points3270/6539/9808. LoRA28 measured steady .435291s/step. Cost .122778 MIGh=.0306945nodeh,30896inputtokens; not interface-effectiveness evidence.; scope: Implementation readiness only, eight twenty-step smokes on 32 dev examples/task [code review: ATT-0022-1 not needed: Pre-review regression and twenty-step smoke after fixture fix; revised E3 analysis]

- EV-0021 [diagnostic, valid] 145tests passed with0failures/errors/skips on e55eea3, including actual ATT0022 smoke rejection, duplicate/aliased paths, incorrect arm/config/schedule/data/order/paramet membership, and complete correctlymapped fixture acceptance. Existing paired bootstrap/selection arithmetic and V0/V1 suite preserved. PBS28s=.007778MIGh=.0019445nodeh; no trainingtokens.; scope: RV0007 formal-input validation regression only [code review: ATT-0023-1 not needed: Regression of RV-0007 validation fix before recheck]

- EV-0022 [diagnostic, valid] Rerun Base full1000/task dev and reload passed. PBS comparison validated formal Base input and exact equality of all2000 prediction records including scores/NLL, dev metric dictionary and split map againstATT0011-2. PIQA.723 BoolQ.680 macro.7015 devNLL2.3374779898760703. Base307s=.085278MIGh=.0213195nodeh; comparator1s=.000278MIGh=.0000695nodeh; no trainingtokens.; scope: Full untrained Base dev rerun and exact E-0002 comparison [code review: ATT-0024-1 covered by RV-0008 pass; ATT-0033-1 not needed: Cheap deterministic equality comparison using reviewed Base reader; no training or selection]

- EV-0023 [diagnostic, valid] E-0004 deterministic readiness checks: 130/130 V0, V1, and added state cases pass. Chunk3 FP32 state max_abs_err r32=0 and r128=3.814697265625e-6 across all three backends, inclusive/strict, S=1/4; chunk1 diagnostics r32=0.17662876844406128 and r128=0.2529714107513428. Capacity boundary 2 accepted and overflow at 1 rejected for all backends. Per-component state corruption controls rejected. Fixed tolerances unchanged. Code review pending; no training or mechanism claim. (supports H-0001@1) [code review: ATT-0034-1 not needed: Deterministic E0 readiness checks; review requested after validation per task contract]

- EV-0024 [diagnostic, valid] Read-only probe succeeded on mg1098/PBS 3760163.opbs: 169 run registry rows, 129 distinct referenced sample paths with zero locally available; nine committed sample archives contain episode/query IDs and predictions but no support facts or question text. Content-level overlap remains unverified. Pinned Qwen tokenizer confirms semicolon ID 26 and space-semicolon ID 2587. Measured PBS 9s = 0.0025 MIG-instance-h = 0.000625 node-h; usage imported from executor observation.; scope: Legacy public assets at a255e814740ff85ab3b4ec4836c88f0a8f802b2d only; no claim about H-0003 or content-level overlap. [code review: ATT-0035-1 not needed: Read-only data schema and tokenizer smoke; no model evaluation]

- EV-0025 [diagnostic, valid] 123 core/model/M-line tests passed in 46.37 seconds on registered GH200 MIG job. Eight-arm M-line Base identity, two-step gradient startup, owner reorder and snapshot isolation, offline controls/oracles and hand-computed read metrics passed. Extended smoke and code review pending.; scope: Pipeline correctness only; no support or challenge to H-0003 [code review: ATT-0036-1 not needed: Deterministic protocol checks and two-step gradient smoke before code review; no costly run]

- EV-0026 [diagnostic, valid] Eight PBS regression tests passed; 80 smoke identities and canonical contents unique; independent regeneration SHA256 matched; 92 seen and 89 unseen single-leading-token names. Legacy pinned source/registry checks establish zero labeled fact-set+question overlap by disjoint relation construction (MSG-0093), not by enumerating legacy episode content. Full dataset generation and Base evaluation pending.; scope: Generator and metric code smoke only; no Base results or H-0003 claim. [code review: ATT-0037-1 not needed: Deterministic generator and metric regression smoke before code review; no Base model evaluation]
- EV-0027 [diagnostic, valid] Eight two-step copy-fixture arms completed, 672 training input tokens, application elapsed 9.26s. Stable step about 0.128-0.172s, 244-328 tokens/s; initialization max_abs=0. Two-step positive-control normal-vs-zero/shuffled criterion not yet met; planned longer smoke follows.; scope: Timing only; two updates are not the positive-control completion test [code review: ATT-0038-1 not needed: Two-step eight-arm timing smoke after 123 passing invariant checks, before review]
- EV-0028 [diagnostic, valid] All eight arms completed 32 smoke updates, all have normal-memory answer NLL below zero and shuffled; 10752 input tokens, 42.53s application elapsed, 303-317 input tokens/s. Validates learnability on tiny copy positive control only.; scope: Two-example copy fixture only; not evidence supporting H-0003 or single-hop generalization [code review: ATT-0040-1 not needed: Task-authorized 32-step tiny copy-fixture smoke before review; measured pilot 9.26s, expected under one minute, no formal training]
- EV-0029 [diagnostic, valid] Full V2 dataset generated twice with matching hashes and 21432 unique episode identities and semantic contents: train16384/calibration2048/dev1000/sealed confirmation2000. 8 tests pass, delimiter and single-token answer checks pass. Train has1381843 input tokens per epoch, 4145529 in three epochs, covering4M per arm. Constant prior dev EM0.015. Legacy overlap0 by authorized disjoint-relation construction proof, not empirical episode enumeration. RV-0010 code review passes with zero findings.; scope: Full generator/data contract validation; no Base or memory-mechanism claim. [code review: ATT-0039-1 not needed: Deterministic data materialization after 8 passing PBS invariants; approximately 30s; no model evaluation]
- EV-0030 [diagnostic, valid] Exact commit 94fb78b integration passed 124 regression tests, eight four-update E16 smoke arms, checkpoint roundtrip and independent train/calibration probe path. 5248 smoke forwarded input tokens; 672 protocol-gradient tokens separately reported. 800-959 tokens/s, training-only V2 extrapolation 2.424 node-hours; two-backend two-LR calibration 0.0588-0.0653 node-hours. Tiny sample is not scientific validation.; scope: E-0005 provisional interface and timing; excludes shuffled address fields pending RV-0011; no H-0003 conclusion [code review: ATT-0041-1 not needed: Coordinator MSG-0106 authorizes provisional E-0005 integration smoke while RV-0010/RV-0011 pending. Four updates per eight arms, plus split probe and regression checks; no formal training.]
- EV-0031 [diagnostic, valid] Reviewed frozen Base evaluator ran with finite full-vocabulary metrics. M32 EM0 NLL9.32865; F32 EM0 NLL5.91752. Measured model load0.849s, M0.287s, F0.164s; used for full-dev walltime sizing, without changing data or protocol.; scope: 32-example dev-prefix timing smoke only; not a task-suitability estimate. [code review: ATT-0042-1 covered by RV-0010 pass]
- EV-0032 [diagnostic, valid] Full dev fixed-protocol Base: M EM0.011 NLL9.2494026874; F EM0.013 NLL5.7063963076; train-selected constant prior EM0.015. M<=prior+0.05 passes;

F>=prior+0.20 fails (required0.215). Per E-0005 outcome_actions this is task-difficulty information for V2 design, not a reason to tune the protocol or a memory-mechanism refutation. Review RV-0010 passed; metrics/predictions complete and finite. Qwen3-0.6B-Base pinned revision, BF16 eager, no training.; scope: Frozen Base dev1000 M/F bounds only. Does not support or challenge H-0003 memory-read hypothesis; no confirmation evaluation. [code review: ATT-0043-1 covered by RV-0010 pass]

- EV-0033 [diagnostic, invalid] Five of six focused checks passed; SoftSlot regression hit exact NLL equality using fused mean CE vs evaluator none-then-mean CE, despite identical logits. Difference 4.77e-7. Corrected test to use defined evaluator reduction without changing production code or tolerance.; scope: Test-reference reduction mismatch; not mechanism evidence [code review: ATT-0044-1 not needed: Focused deterministic regression for RV-0011 medium diagnostic fix before recheck; no training]

- EV-0034 [diagnostic, valid] Six focused checks passed; both backends omit shuffled target-dependent address metrics and preserve entropy/neff. Distinct facts and differing targets with owners[1,0,1] produce identical diagnostic/unhooked logits and defined NLL/EM. Original-target read errors agree within1.27e-7 overall and4.47e-8 shuffled. Normal/permuted address metrics unchanged. PBS3760300.opbs exit0,12s.; scope: RV-0011 diagnostic correction only, no training [code review: ATT-0045-1 not needed: Focused deterministic RV-0011 regression rerun with matching CE reduction; no training]

- EV-0035 [diagnostic, valid] Post-merge regression on commit 3ccde7d: 155 tests passed, zero failures/errors/skips, 46.759s pytest, 49s PBS GH200 MIG; 0.013611 MIG h = 0.00340275 node h. No hypothesis inference. [code review: ATT-0046-1 not needed: post-merge regression]

- EV-0036 [diagnostic, valid] Initial diagnostic smoke: 13 tests passed, then strict E16 original-path NLL check failed. Independent PBS audit reproduced CPU versus CUDA CE reduction difference on identical logits (max 0.00006485 first M batch), zero top1 disagreements in M/F smoke. This is implementation diagnostic only; smoke scientific summaries not produced. Fix preserves original CUDA batch scoring without tolerance changes. [code review: ATT-0047-1 not needed: diagnostic Base evaluation, seconds; reviewed with outputs before delivery; ATT-0048-1 not needed: bounded diagnostic of failed smoke outputs]

- EV-0037 [diagnostic, valid] Fixed 32-calibration smoke: 13 tests passed; diagnostic E16 exactly matches original batch8 predictions and NLL. Prefix-only batch1/8 F top1 agreement 0.96875 and NLL MAD 0.10860 miss stability thresholds; this is timing/smoke evidence, full2048 comparison still required. Application6.635s. [code review: ATT-0049-1 not needed: diagnostic Base evaluation, seconds; reviewed with outputs before delivery]

- EV-0038 [diagnostic, valid] Full calibration2048: diagnostic E16 exactly matches original batch8 EM/NLL. Implementation gate fails: F batch1/8 NLL MAD0.088144 exceeds0.05 (top1 agreement99.2676% passes99%); M passes both. E1 F NLL1.421515 below M9.233152 passes. Per E0007@2 first outcome, investigate implementation and do not use F results for V2 design yet. Descriptive full-vocab EM M0.8301%, F E16 1.4160%, F E1 66.9922%. No H0003 inference; independent code/output review pending. [code review: ATT-0050-1 not needed: diagnostic Base evaluation, seconds; reviewed with outputs before delivery]

- EV-0039 [diagnostic, valid] MSG0135 follow-up after strict BF16 gate failure: true FP32/eager first256 calibration, TF32 off, batch1/8 top1 agreement100% M/F; NLL MAD M3.98e-6

F9.56e-6 versus same-prefix BF16 M0.01325 F0.10252. Supports BF16 numerical batch/padding sensitivity rather than an observed scoring/masking defect in tested path. Full BF16 F absDeltaNLL correlation padding +0.2240, prompt length -0.2165;347 unpadded rows zero difference. Saved12288 records/6 groups audited. Strict original @2 failure retained; MSG0135 diagnostic interpretation allows weak E16 retrieval branch (candidate-b14.209%<50%). No H0003 inference; review pending. [code review: ATT-0051-1 not needed: coordinator MSG0135 authorized bounded FP32 diagnosis; reviewed with outputs before delivery]

- EV-0040 [screening, valid] V1-01 F_out: dev PIQA 0.738, BoolQ 0.818, macroNLL 1.5767698148424294. Complete13077steps/2000040inputtokens/40overshoot,26154examples,388094loss tokens, matching order and parameters, exact reload. Full paired analysis and comparisons in ATT-0052-1. Single-seed screening only.; scope: E-0003@2 seed0 final checkpoint; 1000 dev examples per task; no confirmation [code review: ATT-0025-1 covered by RV-0008 pass]

- EV-0041 [screening, valid] V1-02 F_pre: dev PIQA 0.73, BoolQ 0.811, macroNLL 1.567339824867386. Complete13077steps/2000040inputtokens/40overshoot,26154examples,388094loss tokens, matching order and parameters, exact reload. Full paired analysis and comparisons in ATT-0052-1. Single-seed screening only.; scope: E-0003@2 seed0 final checkpoint; 1000 dev examples per task; no confirmation [code review: ATT-0026-1 covered by RV-0008 pass]

- EV-0042 [screening, valid] V1-03 C_gate: dev PIQA 0.734, BoolQ 0.808, macroNLL 1.5777984531278557. Complete13077steps/2000040inputtokens/40overshoot,26154examples,388094loss tokens, matching order and parameters, exact reload. Full paired analysis and comparisons in ATT-0052-1. Single-seed screening only.; scope: E-0003@2 seed0 final checkpoint; 1000 dev examples per task; no confirmation [code review: ATT-0027-1 covered by RV-0008 pass]

- EV-0043 [screening, valid] V1-04 C_up: dev PIQA 0.736, BoolQ 0.818, macroNLL 1.5652449947787976. Complete13077steps/2000040inputtokens/40overshoot,26154examples,388094loss tokens, matching order and parameters, exact reload. Full paired analysis and comparisons in ATT-0052-1. Single-seed screening only.; scope: E-0003@2 seed0 final checkpoint; 1000 dev examples per task; no confirmation [code review: ATT-0028-1 covered by RV-0008 pass]

- EV-0044 [screening, valid] V1-05 C_dual: dev PIQA 0.739, BoolQ 0.824, macroNLL 1.5619847145143861. Complete13077steps/2000040inputtokens/40overshoot,26154examples,388094loss tokens, matching order and parameters, exact reload. Full paired analysis and comparisons in ATT-0052-1. Single-seed screening only.; scope: E-0003@2 seed0 final checkpoint; 1000 dev examples per task; no confirmation [code review: ATT-0029-1 covered by RV-0008 pass]

- EV-0045 [screening, valid] V1-06 linear: dev PIQA 0.735, BoolQ 0.825, macroNLL 1.5582456389768808. Complete13077steps/2000040inputtokens/40overshoot,26154examples,388094loss tokens, matching order and parameters, exact reload. Full paired analysis and comparisons in ATT-0052-1. Single-seed screening only.; scope: E-0003@2 seed0 final checkpoint; 1000 dev examples per task; no confirmation [code review: ATT-0030-1 covered by RV-0008 pass]

- EV-0046 [screening, valid] V1-07 LoRA4: dev PIQA 0.748, BoolQ 0.836, macroNLL 1.5496020224121514. Complete13077steps/2000040inputtokens/40overshoot,26154examples,388094loss tokens, matching order and parameters, exact reload. Full paired analysis and comparisons in ATT-0052-1. Single-seed screening only.; scope: E-0003@2 seed0 final checkpoint; 1000 dev examples per task; no confirmation [code review: ATT-0031-1 covered by RV-0008 pass]

- EV-0047 [screening, valid] V1-08 LoRA28: dev PIQA 0.749, BoolQ 0.832, macroNLL 1.5535729848258581. Complete13077steps/2000040inputtokens/40overshoot,26154examples,388094loss tokens, matching order and parameters, exact reload. Full paired analysis and comparisons in ATT-0052-1. Single-seed screening only.; scope: E-0003@2 seed0 final checkpoint; 1000 dev examples per task; no confirmation [code review: ATT-0032-1 covered by RV-0008 pass]

- EV-0048 [screening, valid] Reviewed PBS analysis selects Fstar=F_out77.80%, Cstar=C_up77.70%. Best concat Cdual78.15% exceeds Fstar by0.35pp<1pp, hence concat_not_win. Cdual exceeds Cgate by1.05pp (fixed flag true), but nonlinear-vs-linear difference only0.15pp95CI[-.70,1.00]. Cstar trails LoRA4 by1.50pp95CI[-2.40,-.60] and LoRA28 by1.35pp[-2.50,-.20]. All8complete/exactr schedule2,000,040tokens. 10000paired stratified bootstrap seed20261029 descriptive only. Challenges broad H0002 superiority in this setting; no hypothesis-state change or V2 conclusion. Full tables and limitations in T-0003-results.md. (challenges H-0002@1); scope: E-0003@2 tokenwise F-line proxy, seed0, full fixed dev1000/task, final checkpoint; screening only [code review: ATT-0025-1, ATT-0026-1, ATT-0027-1, ATT-0028-1, ATT-0029-1, ATT-0030-1, ATT-0031-1, ATT-0032-1, ATT-0052-1 covered by RV-0008 pass]

- EV-0049 [diagnostic, valid] Verified shared V2 schedule on sealed train: order hash 0d4b55ed8655d91cf0d0532 0.2M=>300 steps,200170 tokens,warmup15; 4M=>6000 steps,4000470 tokens,warmup300. Engineering schedule only, no hypothesis evidence. [code review: ATT-0053-1 not needed: Deterministic schedule preflight only, no training]

- EV-0050 [diagnostic, valid] 146 broad core/model and V2 checks passed in60.43s, source f7c4d2b, including twelve-arm zero-exit/protocol, exact checkpoint reload, fixed donor composition and microbatch mean-CE equivalence. No scientific conclusion. [code review: ATT-0055-1 not needed: Deterministic regression tests before review and calibration]

- EV-0051 [diagnostic, valid] F-line smoke:11 tests passed9.39s;four actual training steps2681tokens, fresh checkpoint reload32calibration+32dev exact per-row match; independent shared sample-order hash and 300/6000step counts match researcher-2. Smoke is not positive-control evidence. Earlier ATT0054-1 cwd failure cost3s and retry24s imported; combined0.001875nodeh. [code review: ATT-0054-2 not needed: Correct cwd for same four-step smoke]

- EV-0052 [diagnostic, valid] All12 arms32step smoke complete; zero-exit logits exact Base, final checkpoint dev predictions identical after reload, controls/curves finite; formal sample schedule independently matches F line. Timing only, not H0003 evidence. [code review: ATT-0056-1 not needed: Twelve-arm bounded smoke before code review;146 regression checks passed]

- EV-0053 [diagnostic, valid] Three300step200170token F calibrations finite/exact reload on2048calibration. LR1e-4/3e-4/1e-3 NLL0.02508849/0.00909726/0.00160201 EM99.414/99.805/99.951percent. Best at upper endpoint requires one3e-3 extension. Calibration signals context task learnability; formal4M and dev remain pending, no H0003 inference. [code review: ATT-0058-1, ATT-0059-1, ATT-0060-1 not needed: E0008 permits 200k calibration after tests and smoke;20tests and real4step smoke passed ATT0057;formal4M awaits review]

- EV-0054 [diagnostic, valid] Reporting-only RV0017 max aggregation fix passed focused regression,1 selected test; no model/metric-selection behavior changed. PBS-generated12-arm forecast1467-1544s per6000step arm including full final eval/curves/reload, requests2201-2316s. Forecast only. [code review: ATT-0068-1 not needed: Focused low-severity RV0017 reporting fix regression and arithmetic timing forecast]

- EV-0055 [diagnostic, valid] Required single upper endpoint3e-3 completed300steps200170tokens,calibrationNI reload. Reviewed PBS selector used calibration only across four candidates and finalized1e-3 (NLL0.00160201). No further extension;formal4M pending. [code review: ATT-0067-1 covered by RV-0018 pass]
- EV-0056 [diagnostic, valid] Six initial backend x LR calibration runs completed successfully,300steps/200170tokens each, all manifests validated. Selection pending registered arithmetic analysis ATT0070-1. Diagnostic LR calibration only; does not support or challenge H0003. [code review: ATT-0061-1, ATT-0062-1, ATT-0063-1, ATT-0064-1, ATT-0065-1, ATT-0066-1 not needed: E0009 explicitly allows calibration after tests/smoke while review RV0017 is pending;146tests and12arm smoke passed; rerun affected calibration if review changes path]
- EV-0057 [diagnostic, valid] Initial grid finite and verified300steps/200170tokens/reload. Kernel NLL5.138646/4.704182/4.627133 and soft5.130321/4.682995/4.627491 for LR3e-4/1e-3/3e-3. Both select upper endpoint0.003; sealed E0009 rule triggers single extension0.01. No H0003 conclusion. [code review: ATT-0070-1 not needed: Deterministic arithmetic analysis of already reviewed training results; no model execution]
- EV-0058 [diagnostic, invalid] T-0010 generator smoke: 9 PBS invariant tests pass; 480 rows independently reproduce with 480 unique identities and contents, zero raw and normalized V2 overlap, verified legacy relation-label disjointness and passing smoke shortcut gates. Not a full dataset or hypothesis test.; scope: E-0011 implementation smoke only; no model evaluation; no H-0005 support/challenge [code review: ATT-0073-1 not needed: deterministic smoke and invariant checks before code review; ATT-0060-1 not needed: E0008 permits 200k calibration after tests and smoke;20tests and real4step smoke passed ATT0057;formal4M awaits review]
- EV-0059 [diagnostic, valid] T-0010 generator smoke: 9 PBS invariant tests pass; 480 rows independently reproduce with 480 unique identities and contents, zero raw and normalized V2 overlap, verified legacy relation-label disjointness and passing smoke shortcut gates. Corrects the result-link recording error in invalid EV-0058.; scope: E-0011 implementation smoke only; no model evaluation; no H-0005 support/challenge [code review: ATT-0073-1 not needed: deterministic smoke and invariant checks before code review]
- EV-0060 [diagnostic, valid] Both prescribed LR0.01 endpoint extensions completed with passing artifacts at300steps/200170tokens. Final selection uses all8 candidates in registered ATT0074-1; no further extensions prescribed. Diagnostic only. [code review: ATT-0071-1, ATT-0072-1 covered by RV-0017 issues, RV-0019 pass]
- EV-0061 [diagnostic, valid] Final sealed-rule choice after single endpoint extension: KernelMatrix LR0.01 normal calibration NLL4.584628655; SoftmaxSlot LR0.003 NLL4.627490530. SoftLR0.01 rawNLL4.606242569 differs by<0.5%, so smallerLR wins. All8 candidates finite. Diagnostic calibration only; no H0003 claim. [code review: ATT-0074-1 not needed: Deterministic final calibration selection using reviewed selector; no model execution]
- EV-0062 [diagnostic, valid] Final reviewed F-line seed0 LoRA6000steps4000470tokens: calibration2047/2048 EM99.95117% vs prior1.90430% passes +20pp gate;dev1000/1000. Nonself2020/2021 and987/987,zero nonselfsubjectcopy. Exact full checkpoint reload;PBS savedoutput audit matches all summaries/counts and curve checkpoints. Single cal error NLL31.875/rank486 explains increased mean NLL0.0155656. Diagnostic context-task learnability only;neither supports nor challenges H0003. Joint E0010 release requires E0009 finite LRs and coordinator

decision. [code review: ATT-0069-1 covered by RV-0018 pass; ATT-0076-1 not needed: Read-only deterministic saved-output audit; no inference or training]

- EV-0063 [diagnostic, valid] E-0011@2 rotation repair: 10 PBS tests pass, inspecting actual terminal edges in all 12 hop/order/noise combinations and the expected 83/84 dev cell sizes; all six hop-by-order ignore-relation scores equal 0.5 in calibration and dev smoke. 480 records reproduce byte-identically with zero V2 overlap and passing shortcut gates.; scope: RV-0020 regression repair; 60 rows per slice, implementation evidence only; no model evaluation or H-0005 support/challenge [code review: ATT-0075-1 not needed: deterministic smoke and regression checks for RV-0020 fix before re-check]

- EV-0064 [diagnostic, valid] Full E-0011@2 T2 preparation passes 10 tests and all data gates: train 18432, calibration 2048, dev 1000, sealed confirmation 2000 and four OOD slices of 1000. All 27480 identities/semantic contents are unique and independently reproduce; zero raw/normalized V2 overlap and legacy label disjointness rechecked. Three training epochs cover 5527605 prompt tokens. No model evaluation yet.; scope: V3 data engineering and structural shortcut gates only; no support/challenge of H-0005 [code review: ATT-0077-1 covered by RV-0022 pass]

- EV-0065 [diagnostic, valid] Base evaluator runtime smoke completes on 60 released V3 dev episodes with finite full-vocabulary scores and hop/order strata, M EM 0 and F EM 0.0333333, before full dev evaluation. Smoke is timing/engineering evidence only.; scope: E-0011 Base evaluator runtime and startup validation, not full difficulty result [code review: ATT-0079-1 covered by RV-0022 pass]

- EV-0066 [diagnostic, pending] T-0011 analysis implementation:21 synthetic PBS regressions passed in2.06s,0 failures/errors/skips. Covers H3/H4 threshold and probe rules, strict V3 near-ties, exact shared bootstrap draws, factorial normalization and formal-artifact corruption rejection. RV-0023 pass0 findings. PBS3762004.opbs on mg1098 exit0,3s walltime=0.000833 MIG-h=0.00020825 node-h. No formal training efficacy or hypothesis conclusion tested.; scope: Source9145309f16231d52bb121118779c986e875951e5; analysis-only synthetic checks; full ten-arm CLI and10000 real-artifact replicates remain pending [code review: ATT-0078-1 not needed: Low-cost synthetic analysis regression before analysis review; no training or scientific decision run]

- EV-0067 [diagnostic, valid] Full 1000-episode V3 Base dev: M EM 0.0, NLL 9.5821453819; F EM 0.009, NLL 5.5676069493, subject-copy 0.079; train-selected dev prior 0.011. M is within prior+5pp overall and for every hop. Per-hop/order outputs retained. Difficulty diagnostic only, no hypothesis support/challenge.; scope: E-0011@2 frozen M/F Base on released dev only; full-vocabulary metrics [code review: ATT-0090-1 covered by RV-0022 pass]

- EV-0068 [diagnostic, valid] Integrated V3 F-line selector smoke: all 26 PBS tests pass, including unchanged V2 defaults/failure boundaries plus V3 release pins, phase-order seeds, sealing and stratum joins. Real V3 LoRA completes four steps/3158 prompt tokens at LR1e-3 and exact fresh-Base adapter reload on 32 calibration/dev rows. Engineering readiness only, not the full 0.2M pilot.; scope: E-0011 pilot selector and checkpoint smoke; no H-0005 support/challenge [code review: ATT-0091-1 not needed: four-step engineering smoke before scoped review of V3 data selector; common F-line engine previously reviewed RV-0018]

- EV-0069 [diagnostic, valid] Fixed V3 F-line LoRA pilot completes 251 steps and 200281 support+question tokens at LR1e-3. Full2048 calibration: EM0.92529296875, NLL0.3153590976.

Hop1 EM0.9297218, hop2 EM0.9604685, hop3 EM0.8856305; hops2-3 pooled EM0.9230769 vs prior0.0205128. Both per-hop and pooled +20pp gates pass. Fresh-Base adapter reload exactly matches all2048 per-episode records and file hashes. No dev or sealed model evaluation in this pilot.; scope: E-0011@2 context-retained T2 learnability under fixed 0.2M pilot; not evidence for H-0005 or permission to start M-line V3 [code review: ATT-0092-1 covered by RV-0022 pass, RV-0024 pass]

- EV-0070 [screening, valid] E-0010@2 locked H3 output: challenged, screening only, seed0 final dev1000. No arm qualifies: max normal-minus-max(zero,shuffled) EM margin is+0.6pp vs required10pp; all target weights are below0.125. Per-arm inputs in V2 order (EM normal/zero/shuffled %, NLL normal/zero/shuffled, target weight):01 [0.9/1.1/1.9;4.427822/9.248186/4.427827;0
[1.7/1.1/1.5;4.448453/9.248186/4.449490;0.059004];03 [1.6/1.1/1.2;4.422075/9.248186/4.423024;0.053784];04
[1.9/1.1/1.3;4.442024/9.249403/4.441777;0.054220];05 [0.8/1.1/1.5;4.427029/9.248186/4.426557;0.058403];06
[1.8/1.1/1.2;4.446569/9.248186/4.447419;0.057738];07 [1.7/1.1/1.1;4.425355/9.248186/4.423455;0.055640];08
[2.0/1.1/1.5;4.445451/9.248186/4.447843;0.058278];09 [2.1/1.1/1.5;4.610641/9.249403/4.609989;0.059095];10
[1.8/1.1/1.9;4.497515/9.248186/4.499404;0.052307]. Qualified0/10; eligible0/10; V3 selection none. Released E-0008 positive control passed, ATT-0069-1. ATT-0093-1/RES-0094 full real-artifact validation and10000 paired bootstrap replicates passed, PBS3762550.opbs on mg1098 exit0/5s. Normal EM0.8-2.1% and value-oracle EM1.1-2.3%: contract qualitative oracle rule points to value-writing/fusion diagnosis, without isolating the cause or ruling out addressing. Confirmation unread. (challenges H-0003@1); scope: E-0010@2 final checkpoints, ten fixed arms including C_dual, dev1000,4M budget/6000steps/4000470tokens each, seed0 screening; not confirmation or a general impossibility claim; coordinator owns conclusion review and hypothesis decision [code review: ATT-0080-1, ATT-0081-1, ATT-0082-1, ATT-0083-1, ATT-0084-1, ATT-0085-1, ATT-0086-1, ATT-0087-1, ATT-0088-1, ATT-0089-1 covered by RV-0019 pass (RV-0023 pending); ATT-0093-1 covered by RV-0023 pass]

- EV-0071 [screening, valid] E-0010@2 locked H4 output inconclusive: SoftmaxSlot better0/4 cells, KernelMatrix better0/4; neither reaches3/4. No supports/challenges flag. Four cell inputs kernel->soft (normal EM%; target weight; n_eff; normalized read error): direct/F_out [0.9->1.6;0.060240->0.053784;9.969244->1.110847;0.165140->2.492829]; direct/C_up [1.7->1.9;0.059004->0.054220;6.274531->1.033493;0.354737->0.488782]; qknorm/F_out [0.8->1.7;0.058403->0.055640;5.618480->2.537960;0.392928->0.289613]; qknorm/C_up [1.8->2.0;0.057738->0.058278;4.044499->5.774581;0.433630->0.245538]. Soft EM is higher in all four cells, but none also has both higher target weight and lower n_eff. C_dual probes excluded. Reader normal-EM factorial effect+0.50pp, paired bootstrap95% interval[-0.05,+1.05]pp; intervals descriptive,10000 draws seed20261029, same episode indices across arms/controls. This does not establish a useful read operator or explain differences by matrix compression.; scope: H-0004 four feature-by-fusion cells only F_out/C_up, final dev1000, seed0 screening; strict simultaneous rule unchanged; hypothesis status remains coordinator-owned [code review: ATT-0080-1, ATT-0081-1, ATT-0082-1, ATT-0083-1, ATT-0084-1, ATT-0085-1, ATT-0086-1, ATT-0087-1, ATT-0089-1 covered by RV-0019 pass (RV-0023 pending); ATT-0093-1 covered by RV-0023 pass]

- EV-0072 [diagnostic, valid] E-0012 L0-L4 measurements complete and independently audited: sixL0 controls96.6278%-100% (min31663/32768; question2048/2048). L1 delimiter-tail direct32648/32768,qknorm30433/32768; last-question supervised address acrossr128/256/512/1024 direct1804-1847/2048,qknorm1684-1756/2048. L2 trained value-tail direct8151-12148/32768,qknorm5556-8615/32768 versus random12609/32768,10539/32768. L3clean-to-clean341-472/2048,read-to-read37-47/2048,clean-to-read38-62/2048 acrossten checkpoints. L4 every trainedarm last-

question argmaxslot0 in1000/1000 dev episodes, target52/1000,same-relation249/1000; key cosine means0.888802-0.999020 with120000pairs/arm; last fusion/residual mean18.6638-1.12365e6 withn1000/arm under stated increment definitions. RV-0027 pass; exact120probe/10L3/14L4 entries; independent counts/baseline/selection/hash/manifest/histogram checks passed; five earlier regressions passed. TotalE0012=1071MIGs=.074375nodeh. Descriptive diagnostic evidence only; no information-absence, causal or route conclusion.; scope: E-0012@1 fixed V2 T1 seed0; train16384 with14336 fit/2048 internal selection, calibration2048, dev1000; L3train4096/calibration2048 ridge1; no confirmation, M-line training, hypothesis update or route selection [code review: ATT-0096-1 covered by RV-0027 pass; ATT-0097-1 not needed: Deterministic independent artifact validator; no extraction, fitting, training, evaluation, split or metric change; measurement core covered by RV-0027]

5 Discussion

证据支持以下几点。第一，V0 侧分支在所测试的恒等式与隔离项上实现正确（diagnostic，D-0003）。第二，在 V1 的固定配方下，gate/up 条件融合没有显示出超过加法写回 1 pp 的优势（screening，D-0004）。第三，V2 的单跳任务在上下文中可学（F 线正对照），但只保留低维记忆时，十组配置都没有产生可用的单跳读取（screening，D-0007）。E-0012 的测量与这样一种解释相容：训练主要通过记忆路径拟合答案先验，没有学会对目标事实寻址。这个解释没有被隔离。测量也不表明信息缺失，因为冻结特征仍能线性解码答案并支持有监督寻址。证据不支持以下推断：记忆机制普遍不可行；任何关于多次读取（H-0005）或读算子选择性（H-0004）的结论；对其他 seed、层、写入位置、事实数或损失的推断。计划 §8.4 允许的温度/增益校准（含重新训练）、S=2 与同布局宽度探针都没有测试，也没有被否定。效度方面的威胁包括：所有训练结果都是单 seed；V1 的学习率只在两个候选中校准；V2 的学习率选择主要反映先验拟合的速度；有监督寻址探针用了额外的目标事实标签；配对区间不含训练 seed 的变异。决策如下：H-0001 记为 supported；H-0002 记为 challenged 并暂停；H-0003 按预期价值停止（D-0008，结论审核 RV-0029），H-0004 记为 inconclusive，H-0005 暂停（D-0009）。按计划 §5.1 与 §14.2，V5 缩减为简单接口对比。针对所观察失败的改动（减少每个 episode 的事实数或课程式训练、单独吸收答案先验的路径、寻址监督）不在允许的探针之内，其中改变事实数需要用户决定。

- D-0001 route:pause — 旧 S-RE 推理资产（writer.pt、sq2 loader 所需文件）不在本文件系统：/work/xg24i002/x10041/LoopLora 目录不存在，只有 GitHub 源码 a255e81 可读。用户 2026-11-01 选择按计划 fallback 跳过冻结 S-RE 诊断（plan §6.1），新方案直接从 Base 实现；不声称旧指标已复现。(reopen if: 用户提供 S-RE 推理资产（writer.pt 及 looplora.llm.sq2.load_inference 所需的归一化 buffers 与配置）的位置，并为该诊断分配 $\leq$1 node·h)

- D-0002 route:continue — V0 实现检查 85/85 通过（ATT-0006-1/RES-0006/EV-0005，GH200 MIG 2g.24gb，容差在适配代码前锁定且未改动）；代码审核 RV-0001 的问题全部修复，RV-0002 通过，两条 low 跟进（FP32 chunk3 状态比较、KernelMatrix 支持容量保护）只约束分块/缓存记忆实验与 V2，不影响 V1 tokenwise 路径。按用户授权的 V0+V1 里程碑进入 V1：先做 E-0002（PIQA/BoolQ 切片与三个家族的学习率校准），再按实测成本生成 E-0003 八组 release。H-0001 的证据状态暂不改为 supported，等 V1 阶段报告时一并做结论审核。V0 实际用量 0.045 MIG instance-h，0 训练 tokens。

- D-0003 route:continue->supported — V0 侧分支实现语义在测试范围内成立（diagnostic，E0）：E-0001 修正后 85/85 项检查通过（RV-0002 pass），E-0004 130/130 项通过（RV-0009 pass）。覆盖陈述中的全部恒等式与隔离项；prefill、分块与逐 token decode 的输出在预先锁定的 BF16 容差内一致；chunk3 FP32 记忆状态在 2e-5 内一致。逐 token（chunk1）记忆状

态的 FP32 差异 0.18–0.25 只作诊断记录，不宣称相等；全 padding 的 query 行无语义约定。结论审核 RV-0014 通过、零发现。范围只是实现正确性，不涉及学习效果或机制价值。下一步：实现作为 V1/V2 的基础继续使用；用到逐 token 缓存记忆的实验前，先查清 chunk1 差异的来源。[conclusion review RV-0014 pass (same_family_unverified)]

- D-0004 route:pause->challenged — 按 E-0003@2 运行前固定的规则（RV-0008 代码审核通过；RV-0015 结论审核通过、零发现）：在 Qwen3-0.6B-Base、F 线、四层 r32 tokenwise proxy、预先锁定的家族校准配方（三族均为 3e-4）、seed 0、每任务 1,000 条 dev、最终 checkpoint 的条件下，最佳 concat（C_dual 78.15%）只比 F*（F_out 77.80%）高 0.35 pp，未达到 +1 pp 的筛选标准，记为 concat 未胜；所选 C*（C_up 77.70%）的观测准确率低于两个 LoRA（79.20%、79.05%）；非线性 proxy 对线性退化组只高 0.15 pp [-0.70, 1.00]。因此 H-0002 在该设置下记为 challenged（screening）。这不排除其他训练 seed、接口专属调参或总体上 $\geq$1 pp 的真实效应（最佳 concat 对 F* 的描述区间上限为 1.35 pp），也不否定有记忆的 concat（H-0003）。按契约，V2 保留 C_dual 作为交互探针，融合用 F*=F_out 与 C*=C_up。此决定不释放 V2 预算。(reopen if: V2 中有记忆的 C*（C_up）或 C_dual 交互探针明显优于 F*（F_out），或用户要求用多个 seed 复核 V1 筛选) [conclusion review RV-0015 pass (same_family_unverified)]

- D-0005 stage:continue_probe — V0+V1 里程碑完成，总用量 11.25 MIG-h $\approx$ 2.81 node·h（预算 200）。V1 没有发现 gate/up 融合相对加法写回的优势（H-0002 暂停），但 V2 的问题（有记忆的单跳读取，H-0003）不由无记忆的 V1 proxy 决定；按 E-0003 契约，V2 融合用 F*=F_out 与 C*=C_up，并保留 C_dual 交互探针。E-0007 显示 E=16 的 F 线对 Base 很难（问题主语复制 96%，限定候选 (b) 14.2%，随机 3.7%；E=1 时 EM 67%），所以建议 V2 分两步：先用训练后的 F 线正对照确认任务在上下文中可学，同时做记忆家族学习率校准；正对照达到预先锁定的水平后，再跑八组析因。V2 是否释放、预算多少由用户决定。

- D-0006 route:pause->inconclusive — E-0010@2 locked H4 rule: SoftmaxSlot better in 0/4 cells and KernelMatrix in 0/4, neither reaches 3/4. With no arm producing a usable episode-memory read (EV-0070), a read-operator comparison cannot separate selectivity from absent memory content; soft EM is higher in all four cells by under 1.2 pp but its target weights are lower in three. No further H-0004 runs until memory reads work. (reopen if: An M-line configuration produces a usable episode-memory read (H-0003 gate met, or the single plan 8.4 probe shows a signal), so the two read operators can be compared on working memory.)

- D-0007 route:diagnose->challenged — E-0010@2 locked rule: 0/10 arms qualify (max EM margin +0.6 pp vs 10 pp; every arm-mean target weight below 0.125 and 1/16; normal and shuffled NLL nearly equal) after the E-0008 positive control passed. Screening only: one seed, released configuration and 4M tokens per arm; not a general impossibility result. The learned-value oracle failure points at values and the downstream path without isolating the cause; LR/schedule, implementation and write-position alternatives are neither shown nor excluded. Per plan 8.4 the next step is E-0012 diagnostic measurement (no automatic mapping), then one preregistered and reviewed probe of at most 4M tokens or a reviewed stop for limited expected value. V3 stays closed. Conclusion review RV-0025 (medium finding on the E-0012 draft mapping) resolved in RV-0026 (pass). [conclusion review RV-0026 pass (same_family_unverified)]

- D-0008 route:stop — Expected-value stop of further spending on the new memory route (M line); H-0003 stays challenged (screening). Basis: E-0010 locked gate 0/10 released arms (EV-0070) and the E-0012 diagnostic measurements (EV-0072; L0 controls 96.6-100%). Observations (single seed, diagnostic): at the last question position the highest-weight slot (argmax of

mixture weights) is slot 0 in 1000/1000 dev episodes for every trained arm, so its target rate equals the slot-0 base rate (52/1000); reads remain weighted mixtures (E-0010 mean n_eff 4.0-10.0 kernel, 1.0-5.8 soft); arm-mean target margins there are negative (soft -2.1 to -16.8, kernel -1247 to -3501); arm-mean fused-increment to layer-14 residual norm ratios are 18.7-608 (1.1e6 for V2-09). Under the L3 ridge probes (4,096 train episodes, calibration held-out) actual-read accuracy is near the prior (1.8-2.3%; clean-to-read 1.9-3.0%) and learned target values give 16.6-23.0%; in the separate L2 softmax probes trained value projections decode the tail no better than random-init projections (17.0-37.1% vs 32.2-38.5%). Frozen layer-14 features give 92.9-99.6% tail decoding and 82.2-90.2% supervised (target-labelled bilinear) addressing top-1 among 16 facts at the last question position across ranks 128-1024, with no consistent gain with rank. Normal and shuffled NLL are nearly equal (4.42-4.61, near the 4.39-nat train answer-prior entropy; zero memory 9.25), while normal is below permuted in V2-03 (-0.18) and V2-10 (-0.11 nats), so outputs are not fully content-independent. Interpretation (not established): predominantly prior-like adaptation through the memory path without demonstrated useful episode-specific retrieval; an optimization cause under answer-only CE is plausible but not isolated; information absence is not shown. Rationale: ten released arms failed; the diagnostics show coupled difficulties; no permitted rescue has demonstrated benefit and their expected value is judged too low to prioritize over the reduced F-line work. Accepted uncertainty: plan 8.2 temperature/gain calibration with a registered restart or paired continuation, S=2 training and wider value/fusion width remain untested and are not falsified; the single plan 8.4 probe (at most 4M tokens) stays available as a reopen path. Scope: one seed, 4M tokens per arm, answer-only CE from Base, layer 14, visible-delimiter writes, E=16, released LRs. Conclusion review RV-0028 (two medium findings) resolved in RV-0029 (pass, two low wording corrections applied here). (reopen if: The user approves a change outside the plan 8.4 probes that targets the observed failure (for example fewer facts per episode or a curriculum, a separate path that absorbs the answer prior, or addressing supervision), or a reason appears to spend the single permitted probe (temperature/gain calibration with restart, S=2, or same-layout width; at most 4M tokens).) [conclusion review RV-0029 issues (cross_model_unverified)]

- D-0009 route:pause — V3/V4 test multi-read gains on a fixed memory that a single read can already use (plan 5.1: V3 only after a usable V2 read). No released V2 arm produced a usable single-hop memory read (EV-0070) and the memory route is stopped (D on H-0003, RV-0029). The T2 chain-end data contract and generator (E-0011, T-0010) are ready and kept for reuse. (reopen if: A working single-hop M-line memory read (H-0003 locked gate met, or a reopened memory route shows the gate signal).)

- D-0010 stage:stop — V2 delivers a stop of the memory route as an expected-value decision (D-0008, RV-0029): no released arm met the locked single-hop gate, diagnostics show a fixed top-ranked slot, read-out increments far above the residual scale and near-prior read decodability, while frozen features remain linearly informative. Per plan 5.1 and 14.2 the project continues with V5 reduced to a simple-interface comparison (V1 tokenwise F_out and C_up against a parameter-matched strong LoRA, F line, PIQA+BoolQ, three seeds, sealed validation opened once). Changes that target the observed failure but lie outside the permitted probes (fewer facts per episode or a curriculum, a separate prior-absorbing path, addressing supervision) are listed for the user; changing facts per episode is the user's decision. [conclusion review RV-0029 issues (cross_model_unverified), RV-0030 issues (cross_model_unverified)]

6 Conclusion

在锁定的配置与预算内（单 seed，筛选级），V1 中 gate/up 条件融合没有优于加法写回，参数相近的 LoRA 更好；V2 中只保留低维记忆时没有产生可用的单跳读取。训练后的模型在答案位置总是把第一个写入的槽位排在最前，读出接近先验水平，而冻结特征仍含有答案信息。记忆路线按预期价值停止，V3/V4 暂停。下一步是缩减后的 V5：在 F 线、PIQA+BoolQ 上，用三个 seed 比较 V1 的 tokenwise 接口（F_out、C_up）与参数匹配的强 LoRA，配置与分析锁定后只开封一次官方 validation。

A Records

- records at data_version 1461; plan DOC-0001@1 (docs/plan.md); constraints docs/constraints.md
- experiments: E-0001@1 (DOC-0003@1), E-0002@1 (DOC-0019@1), E-0003@2 (DOC-0040@2), E-0004@1 (DOC-0047@1), E-0005@1 (DOC-0048@1), E-0006@1 (DOC-0053@1), E-0007@2 (DOC-0063@2), E-0008@1 (DOC-0086@1), E-0009@1 (DOC-0087@1), E-0010@2 (DOC-0083@2), E-0011@2 (DOC-0097@2), E-0012@1 (DOC-0117@1)
- attempts: 106 (7 ended/cancelled, 8 ended/failed, 91 ended/success)
- reviews and advice: RV-0001 answered issues, RV-0002 answered pass, RV-0003 answered issues, RV-0004 answered issues, RV-0005 answered pass, RV-0006 answered pass, RV-0007 answered issues, RV-0008 answered pass, RV-0009 answered pass, RV-0010 answered pass, RV-0011 answered issues, RV-0012 answered pass, RV-0013 answered pass, RV-0014 answered pass, RV-0015 answered pass, RV-0016 answered issues, RV-0017 answered issues, RV-0018 answered pass, RV-0019 answered pass, RV-0020 answered issues, RV-0021 answered pass, RV-0022 answered pass, RV-0023 answered pass, RV-0024 answered pass, RV-0025 answered issues, RV-0026 answered pass, RV-0027 answered pass, RV-0028 answered issues, RV-0029 answered issues, RV-0030 answered issues
- resource use: 0.00583333 mig_instance_hours compute on GH200-MIG-2g.24gb (measured); 0.000138889 node_h compute on GH200 MIG 2g.24gb (measured); 16.8133 mig_instance_hours gpu_mig_instance on GH200 MIG 2g.24gb (measured); 1.2221e+06 input_tokens training on - (measured)